

Force for Free: Target-Error-Augmented State Representation for Contact-Rich Manipulation

anonymous

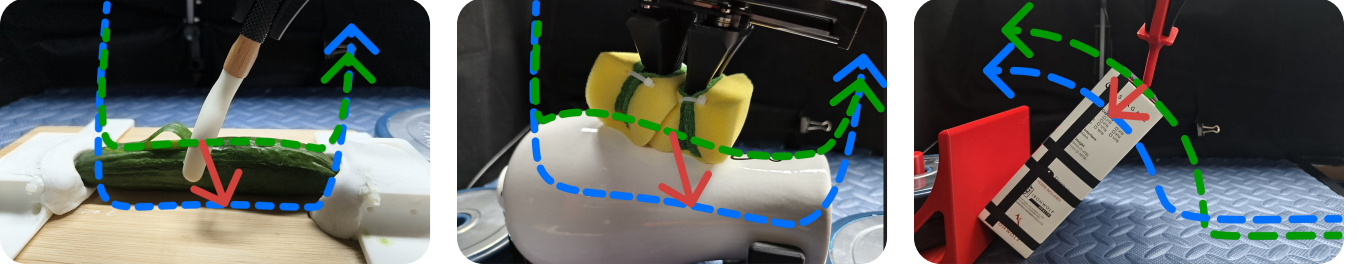


Fig. 1. Depiction of various contact-rich manipulation tasks solved using TEAS in the real world. These include Cucurber Peeling (left), Vase Wiping (middle), and Object Flipping (right). The blue and green trajectories denote the commanded target trajectories and the actual executed trajectories. The red arrows indicate the position tracking errors between them. By leveraging these vectors, our method provides a hardware-free implicit force sensing to enhance visuomotor policies for compliant physical interactions.

Abstract—Contact-rich manipulation remains challenging for visuomotor policies because vision captures geometry well but provide limited information about interaction forces. Existing approaches bridge this gap with dedicated force/tactile sensors or complex force estimators, increasing cost and complexity. We introduce Target-Error-Augmented State (TEAS), a simple drop-in state representation that turns impedance-control tracking residuals into implicit force cues for end-to-end policy learning. The key idea is that, under impedance control, the discrepancy between the commanded target and the actual robot state is not merely a control residual; it is a structured signature of contact. TEAS exploits this by replacing the conventional proprioceptive input with the concatenation of the current state, the commanded target, and their tracking error, without modifying the network architecture or training a separate force model. On a low-cost real robot equipped only with RGB cameras and no force, torque, or tactile sensors, TEAS consistently transforms performance across five contact-rich tasks, achieving approximately 73.9% higher success rates. It remains robust to spatial variations, generalizes to alternative control and teleoperation settings on representative tasks. These results show that control residuals can serve as a practical and scalable alternative to explicit force instrumentation for contact-rich manipulation. Project website: <https://anonymous.4open.science/w/TEAS/>.

I. INTRODUCTION

Contact-rich robotic manipulation has long been recognized as a core challenge in the field of robotics [1]. In recent years, end-to-end visuomotor policies and large Vision-Language-Action (VLA) foundation models have achieved remarkable success in vision-dominant manipulation tasks [2], [3], [4]. Despite these advances, their performance typically deteriorates significantly in contact-rich tasks. Relying solely on visual and standard proprioceptive inputs (e.g., joint positions), these policies often lack the ability to explicitly perceive contact states and accurately maintain contact forces, which are essential for tasks requiring physical interaction with the environment.

To address this limitation, existing methods generally adopt one of two paradigms: incorporating dedicated physical sensors (e.g., 6-axis force-torque or tactile sensors) to acquire explicit force feedback [5], [6], or training algorithmic force-estimation models that infer contact from historical data [7]. However, physical sensors introduce significant hardware costs, integration complexity, and vulnerability to damage, while explicit force-estimation approaches typically demand the construction of high-fidelity simulators, substantially increasing the complexity of the learning pipeline. Both avenues are often cumbersome and difficult to scale in real-world scenarios.

In this work, we propose a remarkably simple yet highly effective method that grants robots the capability to perceive contact without requiring any additional hardware sensors or auxiliary force-estimation models. We introduce **Target-Error-Augmented State (TEAS)**, an enhanced state-representation technique. Unlike current visuomotor policies that feed only real-time proprioception as the system state into the neural network, TEAS augments the state representation by concatenating the desired target command and the actual position tracking error (i.e., tracking residual) together with the standard proprioceptive observation. This approach provides the visuomotor policy with an implicit yet rich “force sense”, leading to a substantial performance improvement in contact-rich environments.

Our method builds upon the inherent tracking residuals naturally generated during impedance control [8]. By incorporating the target commands and the resulting execution errors, the neural network can implicitly infer the interaction forces between the end-effector and the environment. This enables the policy to adjust its output continuously, mirroring the effects of explicit force feedback.

The main contributions of this paper are summarized as follows:

- **Hardware-Free Implicit Force Sensing:** TEAS derives

an implicit force signal from target commands and position tracking errors under impedance control, providing hardware-free contact awareness without expensive and fragile external force/torque sensors.

- **Plug-and-Play State Augmentations:** TEAS is a plug-and-play state augmentation that integrates directly with existing visuomotor architectures without architectural changes, dedicated learning-based and model-based force-estimation modules, enabling simple adoption and deployment.
- **Extensive Real-World Validation:** Across five contact-rich real-world tasks, two policy backbones, and multiple teleoperation/control settings, TEAS achieves near-100% success, strong robustness to spatial perturbations, and compelling evidence of sufficiency through visual and state ablations.

II. RELATED WORK

A. End-to-End Visuomotor Policies for Manipulation

The rise of deep learning has propelled the development of end-to-end visuomotor policies that directly map observations (primarily RGB images and proprioception) to robot control actions. Notable architectures include Action Chunking with Transformers (ACT) [9], which predicts sequences of actions to maintain temporal consistency and reduce compounding errors, and Diffusion Policy (DP) [2], which represents the action distribution as a conditional denoising diffusion process, excelling at modeling multimodal expert behaviors. Recently, large-scale VLA foundation models, such as RT-1 [3], Octo [4], and π_0 [10], have been proposed to leverage massive web-scale datasets to achieve strong generalization. While these models possess unprecedented capabilities in vision-guided manipulation and semantic reasoning, their performance frequently degrades in physical interactions where rich contact dynamics dominate task success. Our work focuses on augmenting these widely-used visuomotor architectures with implicit force awareness without modifying their fundamental network designs.

B. Contact-Rich Robotic Manipulation

Contact-rich manipulation [1] fundamentally involves regulating interaction forces, a classic domain often governed by well-established control theoretic approaches like impedance control [8]. In learning-based methods, acquiring force perception is typically achieved through physical sensors. Many approaches rely on wrist-mounted 6-axis sensors to explicitly measure forces and torques. For example, Adaptive Compliance Policy (ACP) [6], Force-Aware Reactive Policy (FoAR) [11], and Vision-force-fused curriculum learning [12] integrate these measurements into policy learning to adapt compliance. Multimodal approaches, such as Making Sense of Vision and Touch [5], further fuse vision and force readings for enriched state estimation. Meanwhile, other works leverage joint torque sensors (e.g., ForceVLA [13]) to enhance the dexterity of VLA models. The integration of tactile sensors, such as in Reactive Diffusion Policy (RDP) [14], PolyTouch [15], and VTLA [16], has also

enabled robots to capture localized deformation and texture information, advancing fine geometric manipulation.

However, incorporating tactile or force-torque sensors introduces significant hardware costs, integration complexity, and vulnerability to sensor damage during high-impact collisions. To bypass hardware sensors, some works aim to estimate interaction forces algorithmically. For instance, UniFP [7] proposes a unified policy that jointly estimates forces and plans actions through historical states and proprioceptive inputs. Yet, such approaches still necessitate complex simulated environments to explicitly train the force-estimation modules.

Distinct from these methodologies, our proposed TEAS formulation completely circumvents the need for expensive and fragile physical force sensors, nor does it require any additional training in simulated environments. By simply expanding the standard proprioceptive state representation to include position targets and control errors, TEAS unlocks robust performance in contact-rich tasks for visuomotor models in an exceedingly simple and lightweight manner.

III. PRELIMINARIES

A. Impedance Control

In contact-rich tasks, impedance control mode [8] is almost a must choice [5], [17], [7]. Impedance control models the manipulator as a spring-damper system, conferring a compliant interaction with the environment. In our primary experiments, we employ a joint-space impedance PD controller for the operating arm. The control law for the joint torques τ can be generally expressed as:

$$\tau = K_p(q_{target} - q_{curr}) + K_d(\dot{q}_{target} - \dot{q}_{curr}) + \tau_{ff}, \quad (1)$$

where q_{target} and $\dot{q}_{target}$ are the target joint positions and velocities, respectively, and q_{curr} and $\dot{q}_{curr}$ are the current actual joint positions and velocities. K_p and K_d represent the joint stiffness and damping matrices. τ_{ff} denotes the feed-forward torque, typically encompassing gravity and Coriolis compensation.

In our main experiment system, we set the target velocity $\dot{q}_{target}$ to zero. We did not implement gravity compensation nor give any type of feed forward torque, so the τ_{ff} is zero. We can recard the $\dot{q}_{curr}$ as zero since manipulator movement is in low speed. Consequently, the actual torque output by the controller can be simplified as:

$$\tau_{motor} = K_p(q_{target} - q_{curr}). \quad (2)$$

Therefore, a mapping relation between position space and force/torque space naturally exists for impedance control systems.

IV. METHOD

As proved in Eq. (2), the vectors in position space possess force/torque information in a impedance control system. We are going to prove that they contain sufficient force/torque information and add them into the input of visuomotor policies.

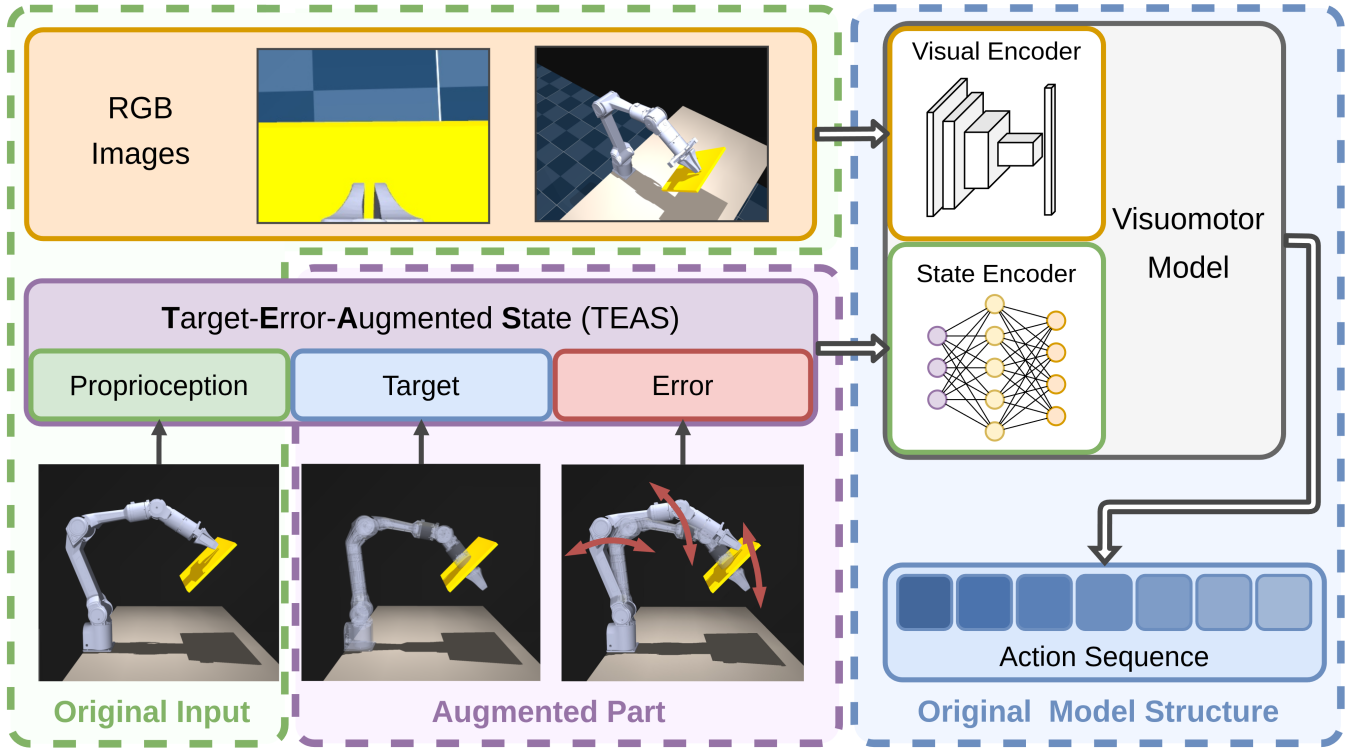


Fig. 3. **Overview of the proposed Target-Error-Augmented State (TEAS).** The core design lies in replacing the standard proprioceptive state representation with a Target-Error-Augmented State Vector. This augmented state is formed by concatenating the standard Proprioception, the commanded Target state, and the tracking Error between them. By explicitly including the tracking error alongside the conventional inputs to the visuomotor model, the policy gains a rich, implicit force sense that significantly boosts performance in contact-rich tasks without requiring physical force sensors.

A. Implicit Force Sensing via Position Tracking Error

According to Eq. (2) and the relationship between Cartesian forces and joint torques derived from the manipulator Jacobian [18], the following mapping can be obtained:

$$e_{pos} \xrightarrow[\text{Parameters}]{\text{Impedance Controller}} \tau \xrightarrow[\text{Kinematics}]{\text{Manipulator}} F_{ext}$$

$q_{current}$ ————

$e_{pos} = q_{target} - q_{curr}$ is the position tracking error, F_{ext} is the external force applied to the robot manipulator. Since F_g for gravity compensation is a constant for a given q_{curr} and $F_{contact} = F_{ext} - F_g$, the mapping can be simplified as:

$$\begin{matrix} e_{pos} \\ q_{current} \end{matrix} \xrightarrow[\text{Robot Manipulator Dynamics}]{\text{Impedance Controller Parameters}} F_{contact}$$

This means the $q_{current}$ and e_{pos} vectors contains all the contact force information. As long as we add them to the observation of the visuomotor policy, it will have the implicit force sensing ability. In practice, we will add q_{target} as well, since it can provide a good stating point for the output action sequence [9]. This might introduce redundant degrees of freedom. But since it cost nothing to add them all, we concatenate them into a state vector and feed it to the existing state encoder of the visuomotor model

B. Target-Error-Augmented State (TEAS)

Typical end-to-end visuomotor policies formulate manipulation as a sequential decision-making process. At time step t , the robot receives an observation O_t and outputs a corresponding action a_t . We define the standard observation space as a tuple:

$$O_t = \{v_t, s_t\}, \quad (3)$$

where v_t denotes the high-dimensional visual representation (e.g., RGB images from multiple camera viewpoints), and s_t represents the low-dimensional proprioceptive state vector. In a standard setup, s_t typically encapsulates only the robot's current joint positions:

$$s_t = q_t \in \mathbb{R}^d, \quad (4)$$

where d is the number of degrees of freedom (DOF) of the manipulator arm.

As mentioned in IV-A, apart from $q_{current}$, we will add q_{target} and e_{pos} into the state vector. Since e_{pos} is directly calculated by $q_{target} - q_{current}$, we will denote the error as Δq in the following section. We introduce the Target-Error-Augmented State (TEAS), denoted as s'_t , which replaces the standard state s_t :

$$s'_t = [q_t, q_{target,t}, \Delta q_t] \in \mathbb{R}^{3d}, \quad (5)$$

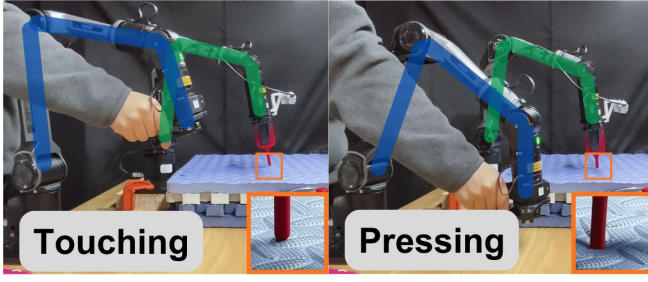


Fig. 4. **Teleoperation Setup.** The operator controls the master arm (blue) to command target positions for the follower arm (green). [Left]: When lightly touching the surface, the target position is maintained slightly above the current position, and the tracking error primarily compensates for gravity. [Right]: When firmly pressing against the surface, the follower arm’s movement is obstructed. Pushing the master arm further down creates a large downward tracking error (target below current) that inherently encodes the applied contact force.

which will form the augmented observation $O'_t = \{v_t, s'_t\}$.

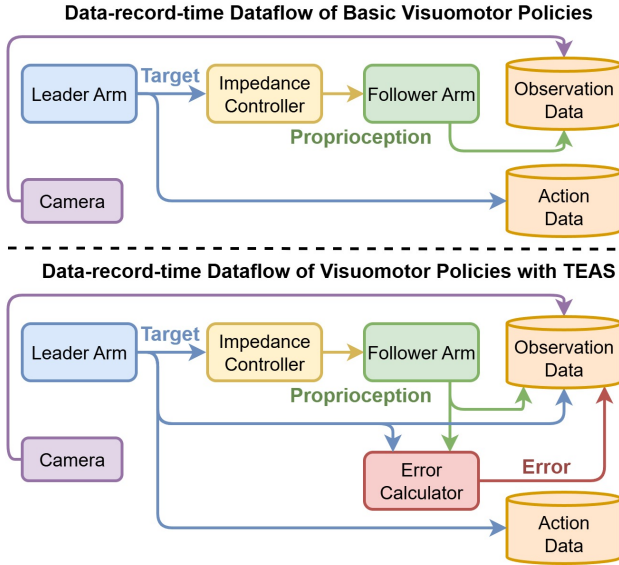


Fig. 5. **Data collection dataflow.** [Top] Conventional dataflow of a standard visuomotor policy, relying solely on vision and proprioception. [Bottom] Dataflow using our proposed method (TEAS). At each time step, the human operator teleoperates the robot via the master arm, providing the target joint position $q_{target,t}$. The tracking error between this target and the follower arm’s actual position q_t inherently encodes the contact force, and is recorded along with visual observations to form the augmented state data.

1) *Data Collection:* At each time step t , the human operator commands the robot using the master arm. As illustrated in Fig. 5, alongside the standard visual observation v_t and the actual follower joint position q_t , we introduce two additional crucial components: the commanded target joint position $q_{target,t}$ (derived directly from the master arm) and the instantaneous tracking error Δq_t , calculated as:

$$\Delta q_t = q_{target,t} - q_t. \quad (6)$$

2) *Network Input Modification and Training:* To apply TEAS, we seamlessly replace s_t with the augmented state

s'_t , formed by the concatenation of the current position, the target position, and the tracking error.

For a d -DOF manipulator (excluding the gripper), the dimensionality of the state vector precisely triples to $3d$. This expansion embeds rich implicit force information without demanding any fundamental architectural changes to the visual state encoder or the core visuomotor policy. The end-to-end model is then trained offline using identical procedures as baseline models, mapping $O'_t \rightarrow a_t$.

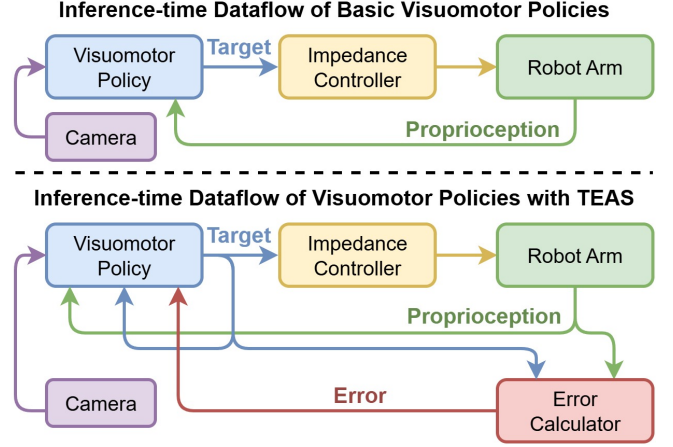


Fig. 6. **Deployment and inference dataflow.** [Top] Inference dataflow of a standard visuomotor policy. [Bottom] Inference dataflow using our proposed method (TEAS). During autonomous deployment, the target command is supplied by an auto-regressive rule using the previously executed action a_{t-1} . This forms the tracking error Δq_t , which is merged with current vision and proprioception into the augmented state s'_t and fed back to the policy.

3) *Deployment and Inference:* During autonomous deployment, the target q_{target} is no longer provided by teleoperation. As illustrated in Fig. 6, for the deploy-time q_{target} input at time step t , we employ a simple, self-sustaining auto-regressive rule: the actual action command dispatched to the robotic arm at the previous time step, denoted as a_{t-1} , is treated as the current target command:

$$q_{target,t} = a_{t-1}. \quad (7)$$

Importantly, a_{t-1} represents the commanded pose practically sent to the hardware for execution, rather than strictly the direct output of a single policy query. This seamlessly accommodates advanced paradigms such as action chunking with temporal ensembling, where the executed action is a fused result of multiple preceding inferences. The actual proprioception q_t is read from the real-time joint encoders, and the tracking error is computed online as $\Delta q_t = a_{t-1} - q_t$. This elegantly bridges the gap between operator-guided demonstration and autonomous execution.

V. EXPERIMENTS: DESIGN AND SETUP

We design experiments to answer the following questions:

- **Q1 (Efficacy):** Can TEAS significantly enhance the performance of visuomotor policies in completing contact-rich tasks?

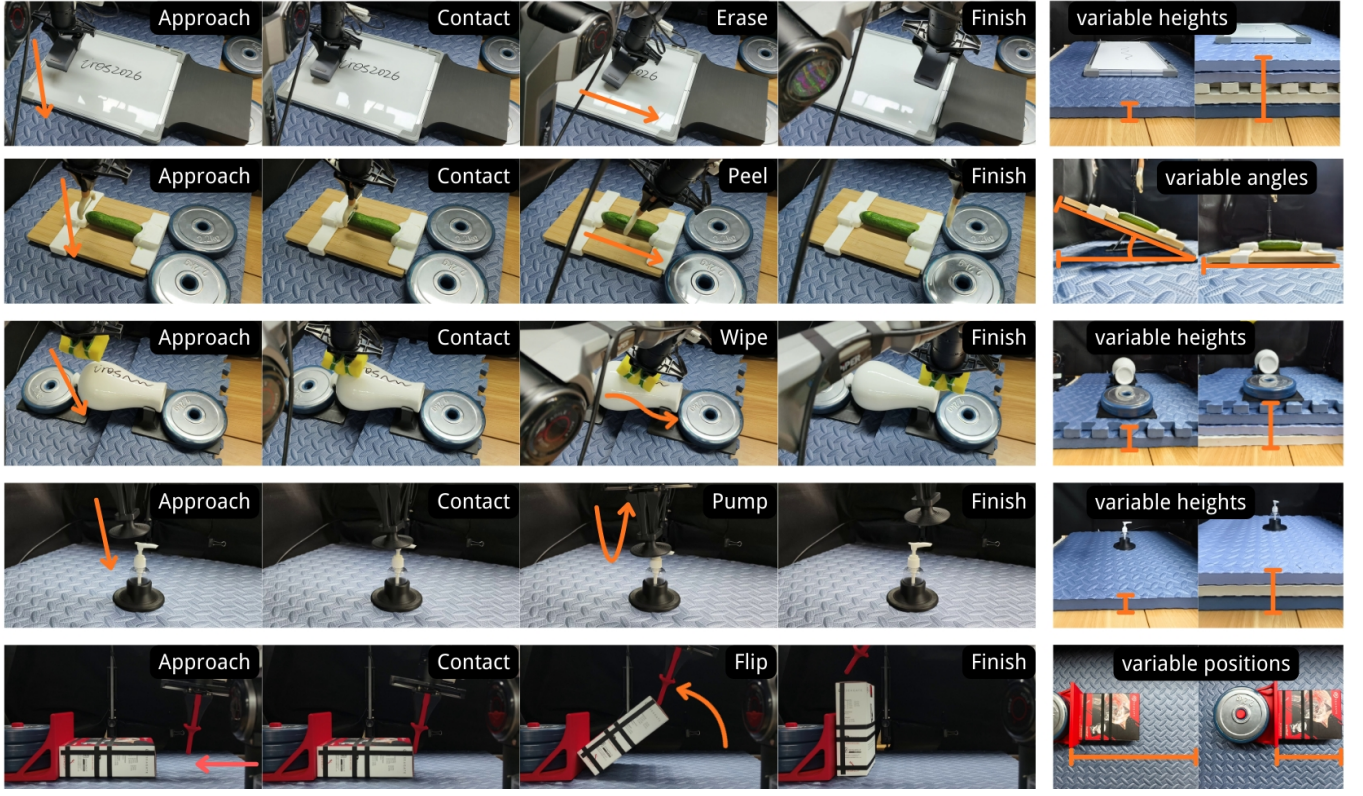


Fig. 7. **Tasks.** We adapted 5 challenging contact-rich tasks from previous works [14], [13], [6], including whiteboard erasing, cucumber peeling, vase wiping, pump pressing, object flipping. These tasks require the policy to sense the contact and apply an appropriate contact force to the manipulated object.

- **Q2 (Compatibility):** Is TEAS applicable across different visuomotor policies?
- **Q3 (Generalization):** Is the TEAS method adaptable to varying teleoperation and control styles?
- **Q4 (Sufficiency):** Does the TEAS possess sufficient information to facilitate force perception?

A. Setup

1) *Hardware:* In all our experiments, we utilize the AgileX PiPER robotic arm as the primary manipulator, which is notably an extremely low-cost platform (approximately \$2000). Crucially, the robot is not equipped with any external force-torque sensors, internal joint torque sensors, or tactile grippers. For visual observations, the system relies entirely on two standard RGB cameras (Intel RealSense D435): one mounted to provide a third-person global view, and another attached to the robot’s wrist. This minimalistic hardware configuration underscores the core advantage of our approach: achieving robust contact-rich manipulation without resorting to expensive and fragile physical force-sensing equipment.

2) *Baselines:* We select two widely adopted end-to-end visuomotor models as our baselines: Action Chunking with Transformers (ACT) [9] and Diffusion Policy (DP) [2]. For the baseline conditions, ACT and DP policies are trained using visual inputs and standard proprioceptive states (i.e., joint positions). For the TEAS conditions, the policies use the augmented state representation s'_t detailed in Section IV. Crucially, the network architectures and hyperparameters

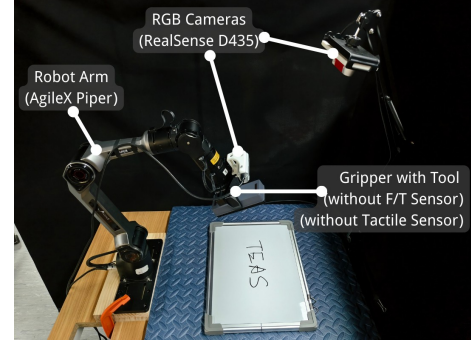


Fig. 8. **Hardware Setup.** We use a classic single arm visuomotor task setup, with a robot arm and two cameras, without any external or internal force sensors.

remain entirely identical between the baseline and TEAS variants.

3) *Tasks:* We evaluate our method across five distinct tasks, as illustrated in Fig. 7. Each of them requires a good sense of contact and to keep an appropriate contact force during the the task period.

- **Whiteboard Erasing.** The robot must sense the initial contact with the whiteboard and maintain a stable wiping force along the surface.
- **Cucumber Peeling.** The robot must perceive contact with the cucumber and carefully regulate the interaction force throughout the peeling process.
- **Vase Wiping.** This task requires the robot to establish

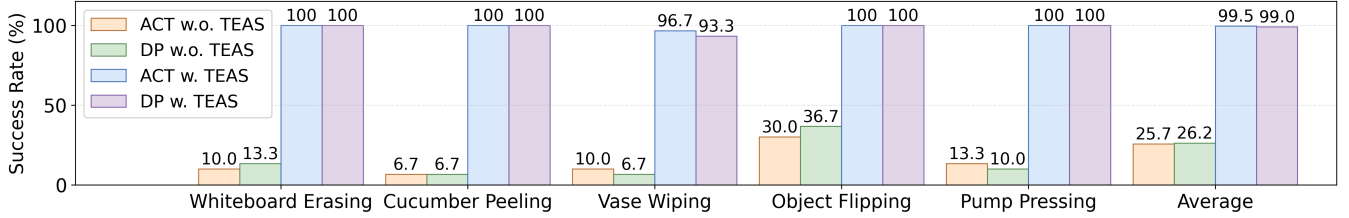


Fig. 9. **Overall Performance.** Success rates (%) of ACT and Diffusion Policy (DP) across five contact-rich tasks, with and without our proposed Target-Error-Augmented State (TEAS). The results show that TEAS significantly improves the performance of both models, achieving nearly 100% success rates on all tasks.

contact and maintain a consistent wiping force while adapting to the contour of the vase.

- **Object Flipping.** The robot applies an appropriate force to one end of the object, requiring a good contact force with a varying direction to assist the flipping motion.
- **Pump Pressing.** The robot must sense contact with the pump head and apply sufficient downward force to press it, followed by a controlled release.

4) *Evaluation Protocols:* For each task, we conduct 30 independent trials to evaluate the performance of our method. A trial is considered successful if the robot completes the task without catastrophic failure. We report the success rate as the primary metric for all evaluated policies.

5) *Implementation Details:* Our implementations of ACT and Diffusion Policy are based on the lerobot [19] library.

VI. EXPERIMENTS: RESULTS

A. Overall Performance

As shown in Fig. 9, we evaluate the overall performance of ACT and Diffusion Policy (DP) across five contact-rich manipulation tasks. The experimental results demonstrate that our proposed TEAS method yields a comprehensive and substantial improvement. While baseline visuomotor policies struggle significantly in these force-sensitive scenarios, augmenting them with TEAS dramatically boosts success rates to nearly 100% on all tasks. This confirms that TEAS effectively provides the crucial force awareness required for robust physical interaction.

To provide a deeper understanding of this performance gap, we present a qualitative vase wiping comparison in Fig. 11. As detailed in the figure, baseline policies fail to adapt their trajectories appropriately when interacting with the object, leading to common failure modes such as excessive force application or premature unit detachment. In contrast, the TEAS-equipped policy intelligently initiates and maintains a stable wiping motion based on the continuously sensed contact.

This ability to effectively perceive and regulate contact stems from the rich information embedded within the position tracking errors. As illustrated in Fig. 10, these tracking errors inherently capture the dynamic vector features of varying contact force conditions throughout the task execution. By directly incorporating this into the state representation, TEAS successfully distills these variations into a functional “implicit force sense”, enabling real-time contact-aware adjustments without the need for delicate external force sensors.

TABLE I
DETAILED TASK SUCCESS RATES.

Method	w.o. variation		w. variation	
	Switch	Erase	Switch	Erase
w.o. TEAS (baseline)	76.7%	69.6%	20.0%	50.0%
w. TEAS (ours)	100.0%	100.0%	100.0%	100.0%
Improvement	+30.4%	+43.8%	+400.0%	+100.0%

(a) Success rates for the whiteboard erasing task.

Method	w.o. variation		w. variation	
	Switch	Peel	Switch	Peel
w.o. TEAS (baseline)	83.3%	68.0%	20.0%	33.3%
w. TEAS (ours)	100.0%	100.0%	100.0%	100.0%
Improvement	+20.0%	+47.1%	+400.0%	+200.0%

(b) Success rates for the cucumber peeling task.

B. Performance Improvement in Easy and Hard Scenarios

To further analyze the policy behavior, we present detailed success rates for whiteboard erasing and cucumber peeling in Table I. These tasks are evaluated under fixed (*w.o. variation*) and varying (*w. variation*) conditions, where variations refer to changes in whiteboard height or cucumber angle. Specifically, *Switch* refers to the critical transition phase where the robotic arm must actively sense the initial contact with the object, accurately halt its downward approach, and smoothly initiate the lateral task motion (wiping or peeling) without either losing contact or applying excessive force. The subsequent phase, *Erase* or *Peel*, evaluates the policy’s ability to complete the remaining trajectory while continuously maintaining an appropriate and stable contact force.

The experimental results highlight the robust advantage provided by implicit force feedback. When spatial setups remain fixed (*w.o. variation*), the baseline visuomotor policy can succeed in a portion of tests by merely memorizing the training trajectories. Crucially, when the target’s height or angle shifts (*w. variation*) and visual observations struggle to accurately capture these novel spatial features, this relative improvement becomes drastically more pronounced. By leveraging implicit force feedback to reliably adapt to environmental changes, TEAS delivers a remarkably larger performance gain over the baseline in varying conditions compared to fixed setups.

C. Generalization on Control modalities

In this experiment, we use Cartesian space impedance control and a joystick teleoperation method. The purpose is

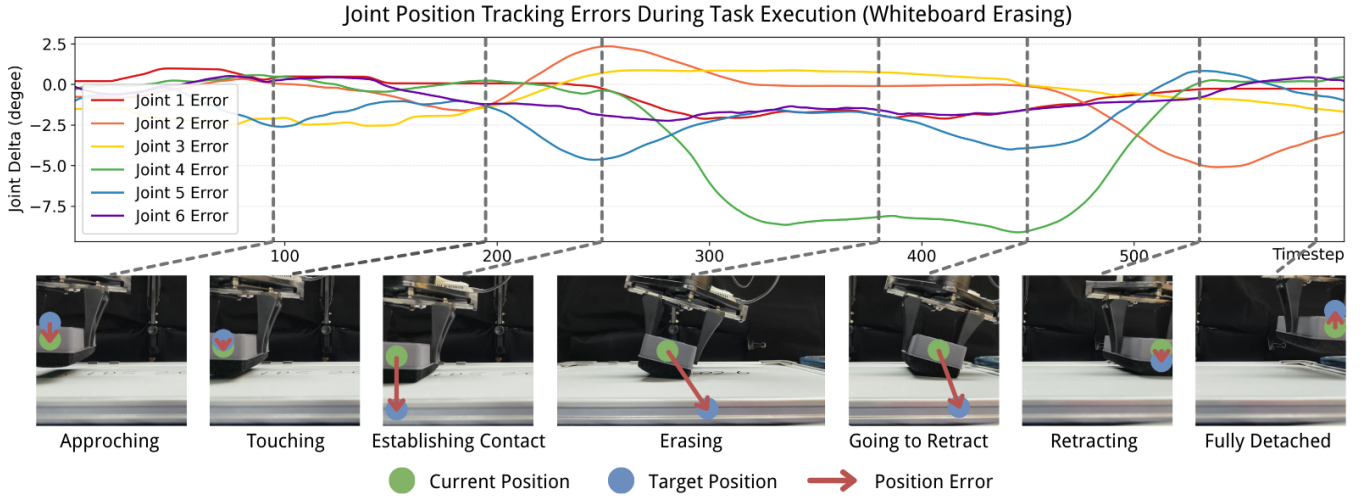


Fig. 10. Joint position tracking errors of the robotic arm under impedance control during a whiteboard erasing task. At different stages of the task, the contact force conditions vary, causing the vector features formed by the joint position errors to change significantly. This demonstrates how the tracking error naturally captures and reflects rich contact information.

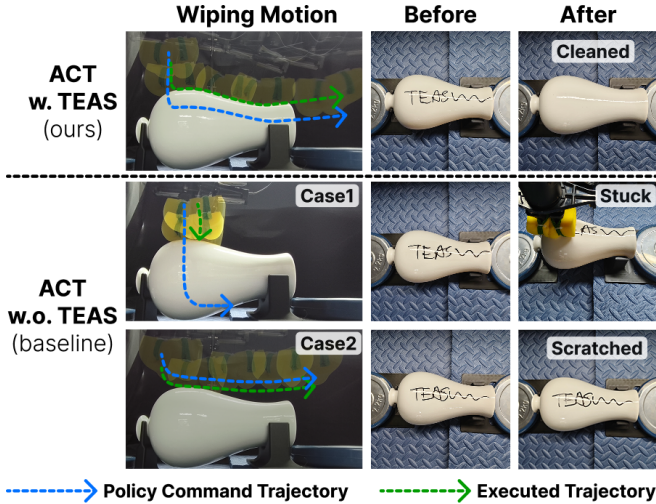


Fig. 11. **Vase wiping comparison.** [Top]: ACT with TEAS. The robot initiates the turning motion only after establishing an appropriate contact force, enabling stable wiping along the vase contour while maintaining consistent contact. [Middle] ACT without TEAS (Case 1). The policy continues dragging the gripper downward, leading to excessive contact force and triggering motor overload protection. [Bottom] ACT without TEAS (Case 2). The turning motion is planned prematurely, resulting in only a slight scratch or even no effective contact with the object. Blue dashed arrows indicate the policy command trajectory, while green dashed arrows denote the executed trajectory.

to test the broad applicability of our method across different styles of impedance control and different teleoperation methods.

For different styles of impedance control, the mapping between the position space and the force/torque space consistently exists. Therefore, the underlying theory of using position error to represent force holds.

In different teleoperation methods, we still have the concepts of target position and current position. We similarly use vectors to represent these positions. Since these two vectors share the exact same dimensions, we can calculate their element-wise difference to obtain the error vector. Concatenating these three vectors constitutes the Target-

Error-Augmented State.

As shown in Table II, our method still achieves a 100% success rate on these representative tasks.

TABLE II
SUCCESS RATES ON REPRESENTATIVE TASKS.

Method	Erasing	Peeling
ACT w.o. TEAS (baseline)	16.7%	10.0%
ACT w. TEAS (ours)	100%	100%

D. Sufficiency Experiment (Visual Ablation)

In this experiment, our purpose is to verify the sufficiency of the Target-Error-Augmented State (TEAS) vector representation for force sensing. To achieve this, we ablate the visual input; therefore, all information obtained by the model comes exclusively from the state vector. Due to the lack of vision to capture the location information of the target objects, we place the whiteboard and cucumber at the same position on the desktop in every trial, but their height and angle will vary.

As summarized in Table III, the baseline policy without visual observation completely fails in these tasks. In contrast, even when entirely deprived of visual input, the TEAS-augmented policy strongly maintains a 100% success rate. This demonstrates that the augmented state vector alone provides sufficient implicit force sensing and spatial awareness for successful task completion.

TABLE III
SUCCESS RATES IN VISUAL ABLATION EXPERIMENTS.

Method	Erasing	Peeling
ACT w.o. TEAS (baseline)	0%	0%
ACT w. TEAS (ours)	100%	100%

E. State Representation Ablation

To understand the contribution of each component in the Target-Error-Augmented State (TEAS) to the policy's

performance, we conduct an ablation study on the state representation. We evaluate the success rate using different combinations of the current joint positions (q).

TABLE IV

SUCCESS RATES IN STATE REPRESENTATION ABLATION EXPERIMENTS.

State Representation	Success Rate
q	10.00%
q_{target}	23.33%
$q + q_{target}$	93.33%
$q + \Delta q$	96.67%
$q_{target} + \Delta q$	93.33%
$q + q_{target} + \Delta q$ (ours)	100.00%

As shown in Table IV, relying solely on the current position q or target position q_{target} results in poor performance. When using the combination of $q + q_{target}$ or $q + \Delta q$, the performance improves significantly since new degrees of freedom are introduced. While the combination of two of the vectors does not show a significant disadvantage compared to TEAS, we suggest to use all the three vectors since each of them stands for information from a certain perspective and it takes no extra effort to add them all.

VII. CONCLUSION

We presented TEAS, a simple yet powerful way to endow standard visuomotor policies with contact awareness by exploiting a signal that impedance-controlled robots already generate: the tracking error between the commanded target and the realized state. Rather than adding force/tactile sensors or training explicit force-estimation modules, TEAS augments the policy input with the current state, the commanded target, and their difference. This small change preserves the original policy architecture while exposing the interaction information needed for contact-rich behavior. In our experiments, TEAS raised average success from 26% to 99% for both ACT and Diffusion Policy, achieving near-saturating performance across five contact-rich tasks and remaining robust under spatial variation, alternative control/teleoperation settings, and representative visual-ablation tests.

Beyond the empirical gains, TEAS also suggests a practical path for future visuomotor systems. By simply replacing the conventional state vector with a target-error-augmented representation, TEAS incurs essentially no additional hardware, modeling, or training cost. As a result, it can be adopted almost for free as the default proprioceptive input when training visuomotor policies under impedance control, exposing implicit contact information already present in the control loop.

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